

Module (12) Overview

Module Introduction

Welcome to Module 12, a comprehensive exploration of machine learning within the realm of computer science. As we journey through this module, we will commence with an introductory lecture that lays the foundation for understanding the field. We will demystify what machine learning is and delve into its expansive scope, ranging from simple rote memory algorithms to complex learning paradigms. This initial session is vital for setting the stage for the advanced topics we will later explore, serving as your gateway into the fascinating and rapidly evolving world of machine learning.

As we proceed, we will dig deeper into specialized topics. Our second lecture introduces Decision Tree Learning, a tool that assists in making informed choices based on various attributes. Following that, we will explore the intricate world of neural networks, including their basic structure and how they emulate human neural pathways. The fourth lecture hones in on perceptrons, which represent the simplest form of neural networks, yet are powerful enough to perform complex tasks when combined. Our fifth lecture ventures into multilayer neural networks, detailing their history, complexities, and the groundbreaking learning rules that have enabled their widespread application. The sixth lecture takes a turn towards unsupervised learning, focusing specifically on self-organizing maps. You will learn how these algorithms can identify patterns and correlations in datasets without the need for explicit labels. Finally, we will conclude the module with a look at advanced topics, including Recurrent Neural Networks and Genetic Algorithms. These specialized techniques open up a multitude of possibilities for machine learning applications, from natural language processing to evolutionary computations.



By the end of this module, you will have a well-rounded understanding of machine learning, equipped with both theoretical knowledge and practical insights. Whether you are a beginner

looking to get started or an experienced student aiming to deepen your expertise, this module offers a comprehensive and in-depth look at a field that is revolutionizing industries across the globe.

Module Objectives

By the end of this module, you should be able to:

- **Explain** the core principles of machine learning, including supervised and unsupervised learning methods.
- **Dissect** the components of a decision tree and understand how it arrives at particular decisions based on attribute values.
- **Analyze** various neural network architectures, including feed-forward and recurrent networks, and discuss their respective strengths and weaknesses.
- **Critically** analyze unsupervised learning techniques like self-organizing maps.

Module Readings

- Russel & Norvig 4th ed: Ch 19.1-19.3, 21.1
- Russel & Norvig 3rd ed: Ch 18.1-18.3, 18.7